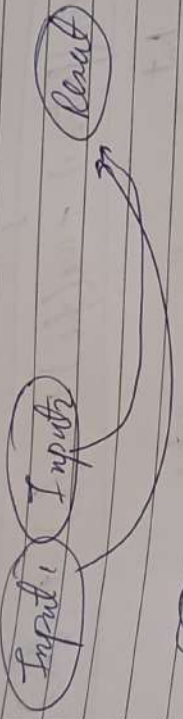


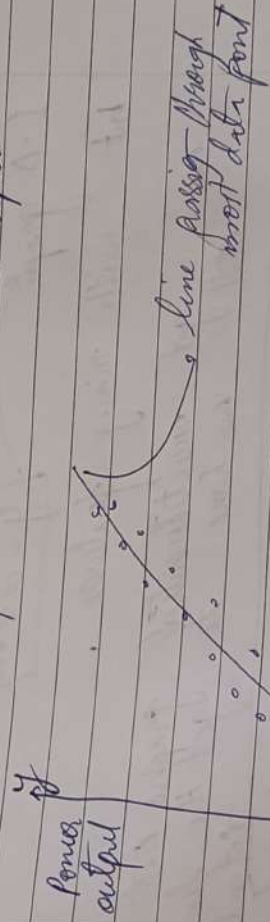
Math for ML (Deep Learning AI)

Week 1

→ Linear Algebra & Machine Learning



(e.g) wind speed → Power output



Wind x

$$y = mx + b$$

finding the best value of m & b

$$y = wx + b$$

weight

bias → const

to find y

Now, if more features are added
which define the output (y) ,
then the equation

$$y = w_1(\text{feature } 1) + w_2(f_2) + w_3(f_3) + \dots + w_n(f_n) + b$$

but in only ~~$y = wx + b$~~

1-D graph

$$y = wx + b$$

but with many features,

we get multidimensional graphs but
still we have to get the linear
relationship

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + b = y$$

different
values
from the
data
set to
get the
 (y) as
close
as possible

$$\begin{array}{l} (1) \quad w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + b = y \\ (2) \quad w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + b = y \\ (3) \quad w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + b = y \\ \vdots \\ (m) \quad w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + b = y \end{array}$$